## EXPERIMENT NO. 07

**LO Mapping:** LO5

**LO5**: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

**Aim:**

To implement monitoring using Prometheus / Grafana / Nagios

**Theory:**

Continuous monitoring is a process to detect, report, respond all the attacks which occur in its infrastructure. Once the application is deployed into the server, the role of continuous monitoring comes in to play. The entire process is all about taking care of the company's infrastructure and respond appropriately.

**Why We Need Nagios tool?**

Here, are the important reasons to use Nagios monitoring tool:

- Detects all types of network or server issues
- Helps you to find the root cause of the problem which allows you to get the permanent solution to the problem
- Active monitoring of your entire infrastructure and business processes
- Allows you to monitor and troubleshoot server performance issues
- Helps you to plan for infrastructure upgrades before outdated systems create failures
- You can maintain the security and availability of the service
- Automatically fix problems in a panic situation
- Nagios is the most popular, open source, powerful monitoring system for any kind of infrastructure. It enables organizations to identify and resolve IT infrastructure problems before they affect critical business processes. Nagios has the capability of monitoring applications, services and entire IT infrastructure.
- NRPE is known as Nagios Remote Plugin Executor. It is designed to execute plugins on remote systems. On Windows hosts, the equivalent remote agent is NSClient++, which listens for Nagios requests and executes local checks (disk, CPU, memory, services, etc.).

- NSClient++ runs as a Windows service on the remote host and waits for requests from the Nagios server (usually over the NRPE protocol, which NSClient++ also supports). When the Nagios server needs to check a resource on that host, it sends a command; NSClient++ executes the check locally and returns the result to Nagios.

**Note on running Nagios on Windows**

Nagios Core is built for Linux and does not install natively on Windows. The standard practice for a Windows machine is to run the Nagios Core server inside a Docker container (using Docker Desktop with the WSL2 backend), and to monitor Windows hosts using NSClient++, which is the Windows counterpart of NRPE. All commands below are Windows PowerShell / Command Prompt commands.

**1 - Pre-requisite: Install Docker Desktop**

Download and install Docker Desktop for Windows and enable the WSL2 based engine. Verify the installation from PowerShell:

```
PS C:\Users\vishal> docker --version
PS C:\Users\vishal> docker version
```

Also make sure the WSL2 feature is enabled:

```
PS C:\Users\vishal> wsl --install
PS C:\Users\vishal> wsl --set-default-version 2
```

**2 - Pull and Run the Nagios Core Docker Image**

Pull a ready-made Nagios Core image and run it as a container, mapping the web port to the host.

```
PS C:\Users\vishal> docker pull jasonrivers/nagios:latest
PS C:\Users\vishal> docker run --name nagios4 -d -p 8080:80 jasonrivers/nagios:latest
```

Verify the container is up and running:

```
PS C:\Users\vishal> docker ps
```

```
C:\Users\Gyaan>docker pull jasonrivers/nagios:latest
latest: Pulling from jasonrivers/nagios
6fe99ef1bed3: Pull complete
a6b90b5bbb15: Pull complete
c42073aaf047: Pull complete
748b632e75e7: Pull complete
7a7447982198: Pull complete
cb259a83ac3d: Pull complete
a7bd6dce6a77: Pull complete
a80db686d641: Pull complete
d003d4303177: Pull complete
fdbaa15b8986: Pull complete
c5701b5e1540: Pull complete
9288c4992679: Pull complete
254354b70ac0: Pull complete
a03720c2c16b: Pull complete
59215b8e47cf: Pull complete
2908452ff193: Pull complete
0d59a81a5608: Pull complete
f47c3af6bf44: Pull complete
dc3a88012a3e: Pull complete
e68fcda12c0c: Pull complete
80b6ab7aef5d: Pull complete
7edde1931fd2: Pull complete
37f75c8146e2: Pull complete
54b9e993e315: Pull complete
e4dc1d4e010c: Pull complete
f2f9a1b3ce48: Pull complete
1413dee23aae: Pull complete
50ea270d34e9: Pull complete
1b9771f55a12: Pull complete
Digest: sha256:fee9cd6a2933e790dde613d70c3a603f96978d65fa6bbc3d617fce807e00f2e2
Status: Downloaded newer image for jasonrivers/nagios:latest
docker.io/jasonrivers/nagios:latest
```

### 3 - Set the Apache/Nagios Admin Password

Set (or reset) the password for the nagiosadmin user inside the running container:

```
PS C:\Users\vishal> docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin
```

Enter and confirm the password when prompted (e.g. 123456).

```
C:\Users\Gyaan>docker run --name nagios4 -d -p 8080:80 jasonrivers/nagios:latest
1238ca84689d7737255d7f43e6c718e07884ffe07e64389d979d36cd91ece6ed
```

### 4 - Access Nagios Web Interface

Access the Nagios web interface from a browser on the Windows host:

```
http://localhost:8080/nagios/
```

Prompting for Authentication Password -

**username:** nagiosadmin

**Password:** 123456 (which you entered while configuration)

```
C:\Users\Gyaan> docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED          STATUS          PORTS                                       NAMES
1238ca84689d   jasonrivers/nagios:latest   "/usr/local/bin/star…"   21 seconds ago   Up 20 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   nagios4
```

### 5 - Install NSClient++ on the Windows Remote Host

On the Windows machine you want to monitor, download and silently install NSClient++ (the Windows equivalent of the NRPE agent):

```
PS C:\Users\vishal> Invoke-WebRequest -Uri
"https://github.com/mickem/nscp/releases/latest/download/NSCP-0.5.2-x64.msi" -
OutFile "NSCP.msi"
PS C:\Users\vishal> Start-Process msiexec.exe -ArgumentList '/i NSCP.msi
ADDLOCAL=ALL MODULES=CheckExternalScripts,NRPEServer /qn' -Wait
```

```
C:\Users\Gyaan>docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
```

### 6 - Configure NSClient++

Edit the NSClient++ configuration file and add the Nagios server's IP address to the allowed hosts list:

```
PS C:\Program Files\NSClient++> notepad nsclient.ini
allowed hosts=127.0.0.1,192.168.1.100
```

Where 192.168.1.100 is your Nagios server (Docker host) IP address.

After saving the file, restart the NSClient++ service so the change takes effect:

```
PS C:\Users\vishal> Restart-Service nscp
```

```
C:\Users\Gyaan>curl.exe -L -o NSCP.msi https://github.com/mickem/nscp/releases/latest/download/NSCP-0.5.2.35-x64.msi
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
  0      0   0      0   0      0      0      0                              0
100      9 100      9   0      0     10      0                              0
```

### 7 - Verify Connectivity from Nagios

From the Nagios server (inside the Docker container), run the NRPE check against the Windows remote host to confirm connectivity. Here 192.168.1.50 is the remote Windows host's IP address.

```
PS C:\Users\vishal> docker exec -it nagios4 /opt/nagios/libexec/check_nrpe -H
192.168.1.50
NRPE v2.15
```

```
C:\Users\Gyaan>docker exec -it nagios4 htpasswd -c /opt/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
```

### 8 - Add the Windows Host in Nagios

Create a configuration file for the new host inside the Nagios container. We also define a service that continuously checks whether the host is reachable (PING check).

```
PS C:\Users\vishal> docker exec -it nagios4 sh -c "vi
/opt/nagios/etc/servers/MyWindowsHost001.cfg"
define host {
    use             windows-server
    host_name       Windows_Host_001
    alias           Windows Host 001
    address         192.168.1.50
    register        1
}
define service{
    host_name               Windows_Host_001
    service_description     PING
    check_command           check_ping!100.0,20%!500.0,60%
    max_check_attempts      2
    check_interval          2
    retry_interval          2
    check_period            24x7
    check_freshness         1
    contact_groups          admins
    notification_interval   2
    notification_period     24x7
    notifications_enabled   1
    register                1
}
```

Verify the configuration file for errors, then restart the Nagios container so the changes take effect:

```
PS C:\Users\vishal> docker exec -it nagios4 /opt/nagios/bin/nagios -v
/opt/nagios/etc/nagios.cfg
PS C:\Users\vishal> docker restart nagios4
```

**9 - Check Host in Nagios Web Interface**

Open the Nagios web interface (http://localhost:8080/nagios/) and check that the new Windows host has been added and is being monitored under Current Status -> Hosts.

**Conclusion:**

The experiment successfully demonstrated the basic implementation of system monitoring using Nagios and NSClient++. NSClient++ was installed and configured on the Windows host, the NRPE service was enabled and running, and the Windows system was configured to allow monitoring requests. The Nagios Docker container was also set up to communicate with the Windows host. Although the final NRPE connection encountered an SSL handshake compatibility issue, the monitoring environment and required services were successfully configured.